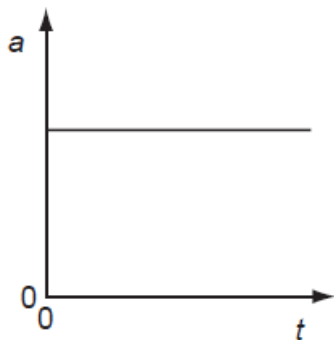


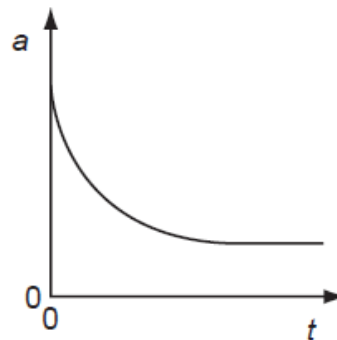
2 A tennis ball is released from rest at the top of a tall building.

Which graph best represents the variation with time t of the acceleration a of the ball as it falls, assuming that the effect of air resistance is not negligible?

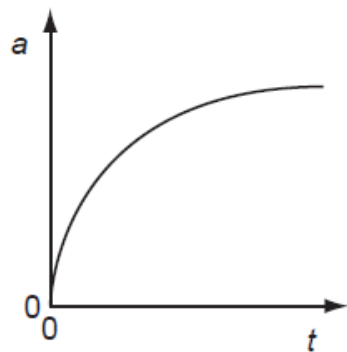
A



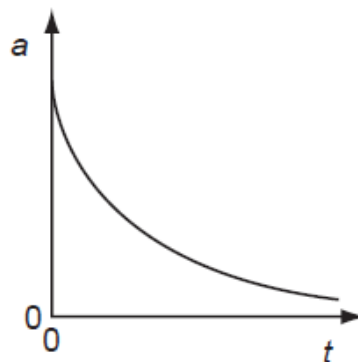
B



C



D



Answer: D

If air resistance is not negligible, acceleration will vary with time. Option A is wrong as it shows acceleration remaining constant.

$F_{\text{net}} = W - \text{Drag force} = ma$. As object falls, velocity increases and drag forces increases. Weight is a constant value. Acceleration will decrease with time and becomes zero when

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	terminal velocity is reached. Option D partly represents this as the graph approaches zero over time.
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2	The acceleration of free fall on the Moon is one sixth of that on Earth. On Earth it takes time t
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